

Recursive Genomic Q-Former: Parameter-Reduction vs. Performance Trade-off

Task details	
Task	cancer_type — 4-way tumor-of-origin classification from bulk gene-expression profiles (TCGA RNA-seq).
Dataset	Unified TCGA cohort: 2,738 samples $\times$ 20,530 genes; top 4,000 highly-variable genes used as input tokens.
Classes	Breast (BRCA, 867), Head-neck (HNSC, 423), Lung (LUNG, 943), Prostate (PRAD, 505).
Split	70/15/15 train/val/test, 523 test samples , seed 42.
Model	Recursive Genomic Q-Former: $d_{\text{model}} = 128$, 256 marker query slots, recursion depth 4.
Metric	Test accuracy and macro-averaged F1.

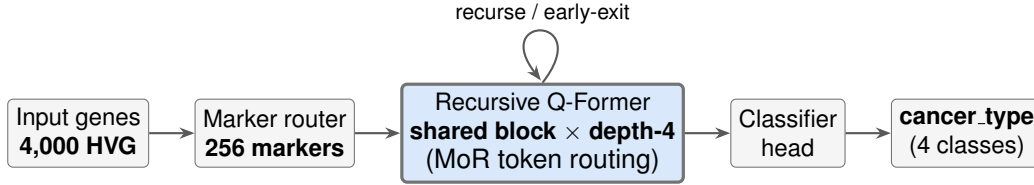


Figure 1: Pipeline. The shared transformer block (blue) is reused across recursion steps; MoR routes tokens to a per-token depth instead of a fixed pass.

All results are on the **cancer_type** classification head (4 classes: BRCA, HNSC, LUNG, PRAD; 523 test samples), with a fixed recursion depth of 4, $d_{\text{model}} = 128$, 256 marker query slots, and seed 42. The “regular” recursive model shares one transformer block across all recursion steps; the comparison models either un-tie those weights (*Independent*) or add token-adaptive routing (*MoR*).

Table 1: Parameters and performance for all three model types. One view of the full trade-off: parameter budget (recurrent-block params, total params, and the independent/shared ratio) alongside all three performance metrics (accuracy, macro-F1, weighted-F1). Sharing the transformer block cuts recurrent params by $\sim 4\times$ for a 0.57-point accuracy cost; Mixture-of-R recursions (MoR) keeps that small budget yet recovers the gap to 0.19 points and is best on macro-F1.

Model	Weights	Routing	Parameters			Performance		
			Recurrent	Total	Ratio	Acc.	Macro-F1	Wtd-F1
Independent stack	untied	none	530,432	1,109,897	$3.99\times$	0.9904	0.9891	0.9904
Recursive (shared)	shared	none	132,992	712,457	$1.00\times$	0.9847	0.9871	0.9847
MoR (token-adaptive)	shared	token	132,992	712,457	$1.00\times$	0.9885	0.9892	0.9885

Read-out. The shared recursive model uses **74.9% fewer** transformer parameters than the independent stack (132,992 vs. 530,432) for a **0.57-point** accuracy cost. Adding MoR token routing

closes that gap to **0.19 points** at the same small parameter budget, and is the only configuration that also reduces FLOPs.

Table 2: Parameter scaling vs. recursion depth. Shared-weight recursion keeps the transformer parameter count essentially flat as depth grows, whereas an independent stack grows linearly. The reduction factor reaches nearly 8 $\times$ at depth 8.

Recursion depth	Shared params	Independent params	Reduction factor
1	132,608	132,608	1.00 $\times$
2	132,736	265,216	2.00 $\times$
4	132,992	530,432	3.99 $\times$
6	133,248	795,648	5.97 $\times$
8	133,504	1,060,864	7.95 $\times$

Table 3: Compute trade-off (depth-4 variants). Early-exit / adaptive recursion lowers the *effective* mean depth and FLOPs per sample relative to a fixed depth-4 pass, at equal parameter count. MoR achieves the lowest compute and the best F1.

Configuration	Transf. params	Mean depth	MFLOPs/sample	Test acc.
Fixed depth-4 (no early exit)	132,480	4.00	268.4	0.9847
Recursive (shared, early-exit)	132,992	2.75	161.5	0.9847
No recursive refine	132,992	2.75	161.5	0.9771
MoR (token-adaptive)	132,992	2.69	159.1	0.9885

Notes. Param ratio in Table 1 is independent/shared transformer parameters. “Transf. params” counts only the recurrent transformer block (the full model incl. embeddings/heads is $\sim 712\text{K}$ for shared variants, $\sim 1.11\text{M}$ for the independent stack). Compute-saving for the early-exit and MoR variants is $\approx 40\%$ of the fixed depth-4 FLOP budget. Token-reduction at the marker stage additionally compresses 4000 input genes to 256 retained markers (93.6% token reduction) at 98.5% accuracy.

Unified results: TCGA cohorts and single-cell (three model variants)

Table 4 collects every dataset for the three model variants in one place. **Top block — TCGA bulk RNA-seq** (2,738 patients, 20,530 genes): tumor-of-origin (`cancer_type`) is modeled *unified* across all four cohorts; the clinical and outcome targets are modeled *cohort-by-cohort* (breast, lung, thyroid, head&neck), so staging and survival are learned without cross-cohort confounding. Because these cohorts are small and imbalanced they use **stratified 5-fold CV** and **macro-F1**, since accuracy there collapses to the majority floor. **Bottom block — single-cell genomap**: Tabula Muris recognition and pancreas integration under the genomap paper’s protocol, reported as **accuracy** (mean $\pm$ std over 5 seeds) with genoNet’s own numbers for reference. The metric therefore differs by block, as labelled in the table.

Table 4: Unified results across domains for the three model types. One table covering both benchmarks for Independent / shared / MoR: **TCGA bulk RNA-seq** (top — `cancer_type` modeled unified, the five clinical / outcome targets modeled separately within each cohort; *macro-F1* %, mean $\pm$ std over stratified 5-fold CV, since accuracy collapses to the majority floor on these small imbalanced cohorts) and **single-cell genomap** (bottom — Tabula Muris recognition and pancreas integration; *accuracy* %, mean $\pm$ std over 5 seeds, paper protocol). Best per row in **bold**; recurrent-block parameters under each model. Metric differs by block, as labelled.

Cohort / dataset	Task	Mean $\pm$ std (metric per block)		
		Independent 530,432 par.	Recursive (shared) 132,992 par.	MoR 132,992 par.
TCGA bulk RNA-seq — macro-F1 %, 5-fold CV				
Unified (all)	cancer_type	97.9 $\pm$ 0.9	96.9 $\pm$ 0.4	96.7 $\pm$ 0.9
Breast (867)	pathologic_stage	26.4 $\pm$ 3.8	25.1 $\pm$ 3.9	26.6 $\pm$ 3.2
	pathologic_T	26.6 $\pm$ 4.4	26.6 $\pm$ 3.7	24.8 $\pm$ 2.3
	pathologic_N	22.8 $\pm$ 1.3	21.2 $\pm$ 2.6	20.7 $\pm$ 4.6
	os_binary	50.2 $\pm$ 5.7	50.4 $\pm$ 3.9	49.3 $\pm$ 5.9
	tumor_status	51.6 $\pm$ 3.4	47.4 $\pm$ 5.0	50.8 $\pm$ 3.0
Lung (943)	pathologic_stage	26.1 $\pm$ 2.5	24.2 $\pm$ 2.9	27.8 $\pm$ 1.6
	pathologic_T	27.3 $\pm$ 2.2	24.7 $\pm$ 1.6	22.4 $\pm$ 3.3
	pathologic_N	22.7 $\pm$ 2.0	24.1 $\pm$ 3.5	24.5 $\pm$ 2.5
	os_binary	46.1 $\pm$ 1.6	46.9 $\pm$ 0.9	46.8 $\pm$ 0.9
	tumor_status	55.0 $\pm$ 2.8	52.6 $\pm$ 5.2	53.1 $\pm$ 3.2
Thyroid (505)	pathologic_stage	25.3 $\pm$ 3.4	24.5 $\pm$ 4.0	24.6 $\pm$ 4.7
	pathologic_T	24.8 $\pm$ 5.9	24.5 $\pm$ 3.3	25.3 $\pm$ 2.4
	pathologic_N	58.0 $\pm$ 3.4	64.8 $\pm$ 2.7	58.9 $\pm$ 4.9
	os_binary	49.2 $\pm$ 2.2	48.7 $\pm$ 0.2	47.2 $\pm$ 3.0
	tumor_status	50.6 $\pm$ 7.3	47.8 $\pm$ 2.1	47.2 $\pm$ 3.2
Head&neck (423)	pathologic_stage	20.4 $\pm$ 4.9	21.8 $\pm$ 3.9	19.2 $\pm$ 2.3
	pathologic_T	20.8 $\pm$ 3.8	20.4 $\pm$ 3.6	19.5 $\pm$ 2.3
	pathologic_N	25.8 $\pm$ 3.8	24.9 $\pm$ 5.5	28.8 $\pm$ 3.9
	os_binary	47.5 $\pm$ 0.2	47.9 $\pm$ 1.6	48.7 $\pm$ 2.3
	tumor_status	50.8 $\pm$ 4.7	51.2 $\pm$ 4.0	50.0 $\pm$ 2.1
Single-cell genomap — accuracy %, 5 seeds (genoNet ref: TM 93, Pancreas 97)				
Genomap	tabula_muris (55)	82.5 $\pm$ 2.7	82.1 $\pm$ 2.2	84.3 $\pm$ 1.1
	pancreas (15)	90.1 $\pm$ 3.0	92.4 $\pm$ 1.3	90.9 $\pm$ 1.6

Notes. Mean $\pm$ std; recurrent-block parameters under each header (total $\sim$ 1.11M Independent; $\sim$ 712K shared and MoR). **TCGA block**: macro-F1 % over stratified 5-fold CV (every sample tested once); `cancer_type` is the unified four-

cohort model (results_cohorts_cv/*/unified/), the per-cohort results from results_cohorts_cv/*/(cohort)/. Only tumor-of-origin (~97%) and thyroid node status (~59–65%) carry strong signal; staging targets hover near the 25% chance level and binary outcomes near 50%, so fine-grained staging is not recoverable from bulk expression at these cohort sizes. **Single-cell block:** accuracy % over 5 seeds under the genomap protocol (70/30 / integration split, lr 0.001, wd 10^{-5} , batch 128, 150 epochs; results_singlecell_ms/*/); genomap's genoNet CNN (refs 93% TM, Suppl. S4; 97% Segerstolpe pancreas, Fig. 7c) is stronger, but the *same* 0.7M-parameter model transfers unchanged from bulk to single-cell (MoR best on TM, shared best on pancreas). No single variant dominates across either block.